Elias-Fano coding

Advanced Data Structures – Summer 2026 – Lecture 2

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2026-04-27 Mon 14:00

Previous slides: Bit vectors, Rank & Select

- Bit vector: storing bits
- Rank: counting 1 bits
- Select: finding 1 bits

Goal: Storing lists of integers

How do we efficiently store a list of integers

$$0 \leq x_0 \leq x_1 \leq \cdots \leq x_{n-1} < U.$$

Theorem 1 Lower bound.

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For $n = o(\sqrt{U})$ and $n \to \infty$,

$$\frac{1}{n} \log_2 \binom{U+n-1}{n} \to \log_2 \left(e \cdot \frac{U}{n} \right).$$

Storing many small integers ($n \gg U$)

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- Lower bound for $n = u$:

$$\log_2 \binom{U+n-1}{n} = \log_2 \binom{2n-1}{n} = \log_2 \left(\Theta(2^{2n}/\sqrt{n})\right) \approx 2n$$

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  - $\log_2 n - \log_2 e$ bits/key wasted!

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Consecutive elements share the $\approx \log_2 n$ high bits!

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Theorem 2 Elias-Fano (EF) coding.

A list of n integers $0 \leq x_0 \leq x_1 \leq \cdots \leq x_{n-1} < U$ can be stored in

$$n(\log_2(U/n) + 2)$$

bits while supporting $O(1)$ access time.

This is only $2 - \log_2(e) = 0.5573$ bits/key away from the lower bound when $n = o(\sqrt{U})$!

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- Using the $n \approx U$ case!

Elias-Fano coding

Definition 1.

Split each integer into a $\ell = \lfloor \log(U/n) \rfloor$-bit *low* part and $(\log U - \ell)$-bit *high* part.

- Store all low bits explicitly in $n\ell$ bits.

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- Write how often each high part occurs in unary.
  - $\lceil U/2^\ell \rceil$ high parts with

$$n \leq \lceil U/2^\ell \rceil < 2n.$$

  - In total $n + \lceil U/2^\ell \rceil \leq 3n$ bits.
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- Store all low bits explicitly in $n\ell$ bits.
- Thus, space usage below $n\ell + 3n = n(\lfloor \log(U/n) \rfloor + 3)$
- More refined analysis shows that actually: $n\lfloor \log(U/n) \rfloor + n + \lceil U/2^\ell \rceil \leq n(\log(U/n) + 2)$.

Interpretations of the high-bit encoding

- Each input number corresponds to a 1.
- Each "bucket" for the high parts corresponds to a 0.
- 0 separates the unary count in number of 1 for each high part.
- The number of 0 directly before a 1 is the jump the high part makes from the previous value.
- The total number of 0s before a 1 is the value of the high part.

Querying Elias-Fano

Definition 2.

To get x_i:

- Read the i'th group of ℓ low bits.

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To get x_i:

- Get the high part of the i'th value:
  - Count zeros before the i'th 1 (counting from 0):

$$\mathsf{high}(i) = \mathsf{select}_1(i) - i$$

- Read the i'th group of ℓ low bits.

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- For *strictly increasing* integers: replace x_i by $x_i - i$.
- To store a non-sorted list with small sum: store prefix sums.

Further reading / sources

- Giulio Pibiri's notes on Elias-Fano
- Modern paper reviving/applying it: (Vigna 2013)
- Original papers: (Elias 1974; Fano 1971)
- Dynamic version: (Pibiri and Venturini 2017)
- `sux::EliasFano` implementation

Bibliography

Elias, Peter. 1974. “Efficient Storage and Retrieval by Content and Address of Static Files.” *Journal of the Acm* 21 (2): 246–60. https://doi.org/10.1145/321812.321820.

Fano, R.M. 1971. *On the Number of Bits Required to Implement an Associative Memory*. Memorandum 61, Computation Structures Group. MIT Project MAC Computer Structures Group. https://books.google.ch/books?id=07DeGwAACAAJ.

Pibiri, Giulio Ermanno, and Rossano Venturini. 2017. “Dynamic Elias-Fano Representation.” In *Lipics, Volume 78, Cpm 2017*, 78:30:1–30:14. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. https://doi.org/10.4230/LIPICS.CPM.2017.30.

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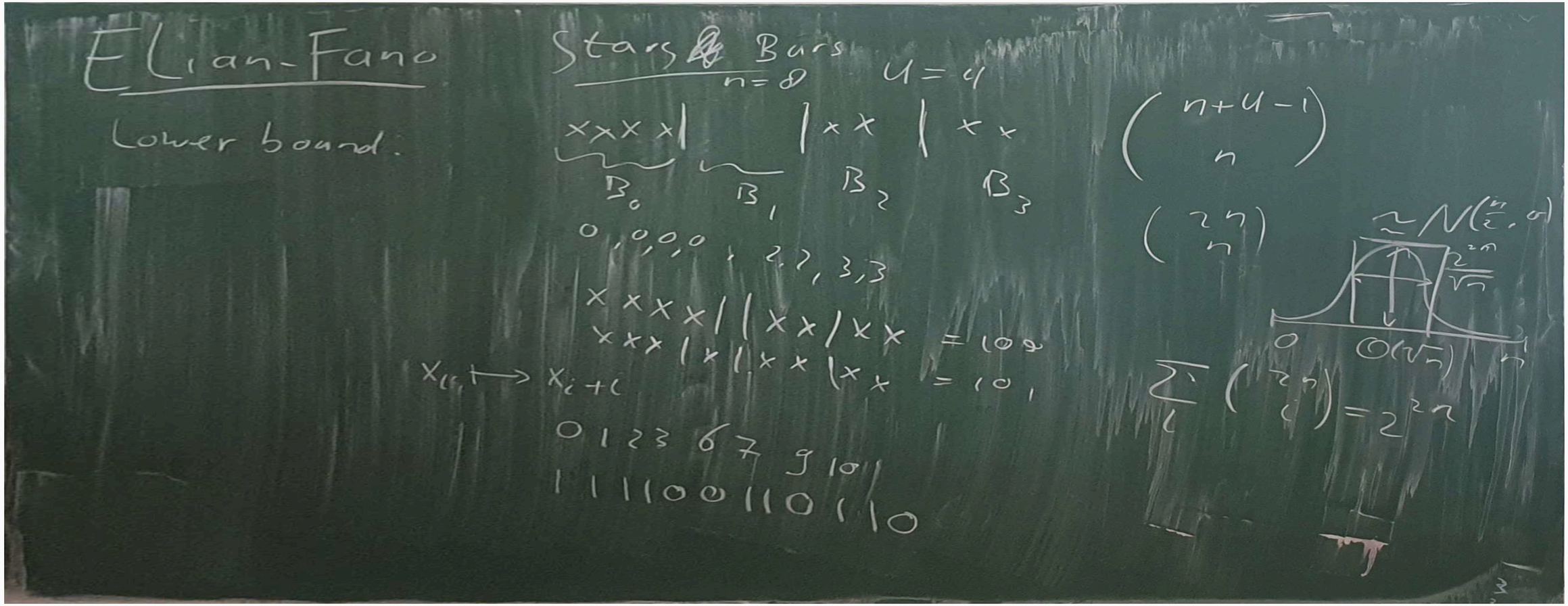
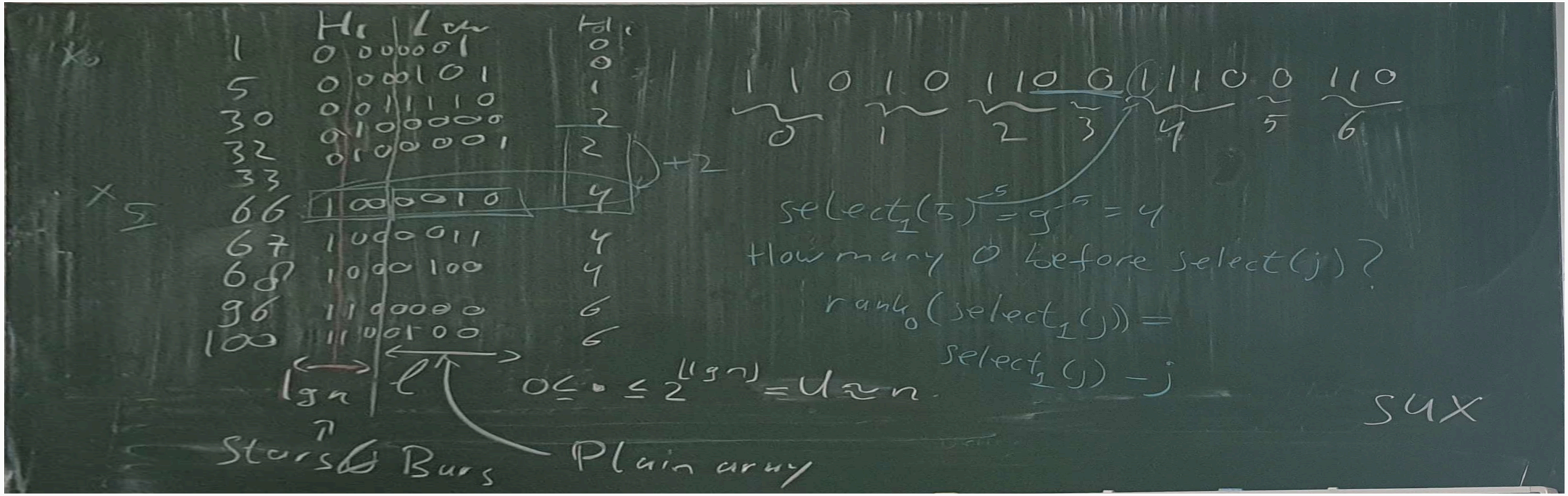
Possible exam questions

- What is the core idea of Elias-Fano coding?
- What is the stars & bars technique? Relate it to a binomial coefficient.
- How may "high" bits are there? Why does this make sense?
- What is the space of the Elias-Fano encoding? Is it far from the lower bound?
- What are some interpretations of the bits encoding the high parts?
- How does one recover x_i, i.e., query the Elias-Fano coded list?

Blackboard: Bit vector & Rank



Blackboard: Select & Darray